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StandDao

On-chain reputation and soulbound credentials

Built on BOT Chain testnet (chain ID 968)

Prepared for the BOT Chain Africa Builder Challenge

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Contents

1. Abstract
2. The Problem
3. The StandDao Approach
4. Core Concepts
5. How It Works
6. Worked Example
7. Smart Contract Design
8. Governance and the Path to a DAO
9. Application Layer
10. Deployment on BOT Chain
11. Security Considerations and Limitations
12. Roadmap
13. Conclusion

1. Abstract

StandDao is an on-chain reputation and credential protocol for BOT Chain. Approved issuers award non-transferable (soulbound) credentials to wallet addresses, each worth a number of reputation points. Holders with sufficient standing can endorse others, and any smart contract can query a wallet's score to gate access to pools, votes, bounties or other features. Credentials can be earned and revoked, but never bought, sold or transferred.

This paper explains the problem StandDao addresses, how credentials and scores work, the contract architecture, the current administrative model and its limits, and the intended path toward community governance.

2. The Problem

On-chain communities need to know who to trust, but an address by itself says nothing about a person's skills, history or conduct. Existing solutions each fall short for community use.

- **Tokens are for sale.** Governance tokens and ordinary NFTs can be bought, so holding one signals wealth rather than contribution.
- **Reputation is scattered.** A developer's completed bounties, a lender's repayment record and a moderator's community work live in separate silos that other applications cannot read.
- **Identity systems are heavy.** Full decentralized identity stacks demand complexity that small communities rarely need.

In African builder communities this gap is felt strongly. Talented people are often unknown to distant counterparties, and a portable, verifiable record of completed work can open doors to funding, jobs and lending that would otherwise depend on personal introductions.

3. The StandDao Approach

- **Soulbound by construction.** The contract contains no transfer function at all. A credential stays with the wallet that earned it.
- **Issuers carry the trust.** Credentials are only as meaningful as the organisation that issued them, so each credential records its issuer. Applications can choose which issuers they respect.
- **A single, simple score.** Each credential adds points and each endorsement adds one, giving applications a number they can check in one call.
- **Composable gating.** A view function lets any contract require a minimum score with one line of code.

4. Core Concepts

Concept	Description
Credential type	A named achievement defined by the owner, such as "Verified Solidity Dev", with a point value.
Issuer	An address approved by the owner to award credentials.
Credential	A record that a specific wallet holds a specific type, with its issuer, timestamp and revoked flag.
Endorsement	A one-point vote of confidence from one wallet to another, available to wallets above a score threshold.
Score	The sum of points from active credentials plus endorsement points.

5. How It Works

Defining credentials

The owner adds credential types, each with a name and a point value. Types are numbered in the order they are created. A type cannot be deleted in this version.

Issuing

An approved issuer calls `issue` with a recipient and a type. The contract rejects the call if the caller is not an issuer, the type does not exist, or the recipient already holds an active credential of that type. On success it records the credential and adds its points to the recipient's credential score.

Revoking

The original issuer of a credential, or the owner, can revoke it. Revocation marks the record as revoked, frees the type so it can be re-issued, and subtracts its points from the holder's score. The history remains visible on-chain.

Endorsing

Any wallet with a score at or above the endorsement threshold (10 by default) can endorse another wallet, once per pair, but never itself. Each endorsement adds one point to the subject. Endorsements cannot be withdrawn in this version.

Scoring and gating

```
score(wallet) = sum(points of active credentials) + endorsementPoints(wallet)

// in any other contract
require(rep.meetsScore(msg.sender, 20), "Reputation too low");
```

Because the score is a view function, gating costs the caller nothing extra and works inside any contract on the chain.

6. Worked Example

Step	Effect on the wallet's score
Builders Guild issues "Verified Solidity Dev" (20 points)	20
Bounty Board issues "Completed 5 Bounties" (15 points)	35
Three wallets above the threshold each endorse	38
The Bounty Board revokes its credential	23
A lending pool requires a score of 20 to borrow	Still eligible

7. Smart Contract Design

The protocol is a single contract, `SoulboundReputation`. It is intentionally not a full ERC-721 token: it has no transfer, approval or metadata functions. It follows the ERC-5192 convention by exposing `locked()`, which always returns true, so tooling that understands locked tokens recognises it.

Function	Access	Purpose
addType	Owner	Define a credential type and its points.
setIssuer	Owner	Approve or remove an issuer.
setEndorseThreshold	Owner	Change the score needed to endorse.
issue	Approved issuers	Award a credential.
revoke	Original issuer or owner	Revoke a credential.
endorse	Wallets above threshold	Add one point to another wallet.
scoreOf / meetsScore	View	Read a score or test it against a minimum.
getCredential / getType / counts	View	Read records for interfaces.

Events and safety patterns

- Events `TypeAdded`, `Issued`, `Revoked` and `Endorsed` allow full reconstruction of any wallet's history.
- Duplicate credentials are prevented per type, and revoked credentials free the type so a holder can be re-issued after a correction.
- Score arithmetic uses Solidity 0.8 checked math and a revocation always subtracts exactly the points that were added.

8. Governance and the Path to a DAO

The name StandDao describes where the protocol is heading, not where version 1 stands. In this version a single owner address administers the system: it defines credential types, approves issuers and sets the endorsement threshold. That is a pragmatic way to launch a community with a small set of founding issuers, but it is a point of centralisation and the document states it plainly.

The intended direction is to hand these powers to the community, in stages: first a multisig of founding issuers, then on-chain voting among credential holders weighted by standing, so that the people whose credentials the network trusts also decide who may issue them.

9. Application Layer

The reference application is a static site with no backend. It provides a landing page, a profile page that shows any wallet's score, credentials, issuers and dates, with an endorse button, and an issuer console where the owner defines types and issuers and approved issuers award credentials. Wallet connection prompts users to switch to or add BOT Chain testnet, and all reads come from the public RPC endpoint.

10. Deployment on BOT Chain

Parameter	Value
Network	BOT Chain testnet
Chain ID	968
RPC endpoint	https://rpc.bohr.life
Block explorer	https://scan.bohr.life
Contract address	Published at deployment
Language / compiler	Solidity 0.8.24 (checked on solc 0.8.26)

11. Security Considerations and Limitations

- **Not audited.** The contract is a reference implementation and has not been independently reviewed.
- **Owner trust.** The owner can add types and issuers and can revoke any credential. The contract has no ownership-transfer function in this version, so the deploying key must be protected, ideally as a multisig.
- **Issuer quality.** A malicious or careless issuer can award credentials freely. Applications should whitelist the issuers whose credentials they accept, which is possible because every credential records its issuer.
- **Endorsement rings.** Groups of wallets could endorse each other to inflate scores. The threshold and one-endorsement-per-pair rule limit this but do not eliminate it. Applications with high stakes should weight credentials more heavily than endorsements.

- **Privacy.** Credentials are public and permanently linked to an address. Users should consider this before accepting a credential, and issuers should avoid recording sensitive personal information.
- **No expiry.** Credentials do not expire, so a record of a skill does not decay with time unless an issuer revokes it.

12. Roadmap

Stage	Focus
Next	Independent audit; ownership transfer and two-step handover; verified source on the explorer; indexed profile pages.
Then	Multisig administration, credential expiry, ability to retract endorsements, and the option to deactivate credential types.
Later	Signed off-chain attestations (EIP-712) that holders can claim on-chain; full ERC-721 metadata compatibility; token-weighted or standing-weighted DAO governance over issuers.

13. Conclusion

Trust in an open network cannot be bought if it is to mean anything. StandDao records what a wallet has done, who vouched for it, and lets any application act on that record in a single call. Starting with a small set of trusted issuers and moving toward community governance, it aims to give builders across Africa and beyond a reputation that travels with them.

Disclaimer

> This document describes software that is experimental, deployed on a test network and has not been independently audited. Nothing here is financial, legal or investment advice, and nothing here promises returns or the availability of any feature. Test-network tokens have no monetary value. Roadmap items are intentions, not commitments.